

Invariants in Complex Dynamics*

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Abstract

Using Newton's method for complex polynomials as a guiding thread, this lecture aims to show the different behaviors that the orbits of a dynamical system generated by the iteration of an analytical function of the complex plane can have. We will see how these orbits group into invariant sets with very varied dynamics, separated by fractal boundaries with remarkable topological and dynamical properties.

1 Newton's method: the origins

Newton's method is probably the best-known and oldest iterative method that can be found in mathematics. This method is used to calculate approximate real or complex solutions of an equation

$$f(z) = 0,$$

where f must be a differentiable function and z is a real or complex variable. Then we construct the function

$$N_f(z) = z - \frac{f(z)}{f'(z)}$$

and observe that $f(\alpha) = 0$ if and only if $N_f(\alpha) = \alpha$.

The local study of Newton's method dates back to the year 1689, when Joseph Raphson proved the following theorem.

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Theorem 1.1 (Local Study. Raphson, 1689). *If $f(\alpha) = 0$ and z_0 is close enough to α , then the sequence*

$$z_0, z_1 = N_f(z_0), z_2 = N_f(z_1), \dots, z_n = N_f(z_{n-1}), \dots$$

converges to α .

The point z_0 is called an initial condition and the sequence $\{z_n = N_f^n(z_0)\}_{n \in \mathbb{N}}$ is known as the orbit of z_0 . Thus, the theorem tells us that any root α of f has a neighborhood of initial conditions whose orbit converges to α , or in other words, for which the method “works” (see Figure 1).

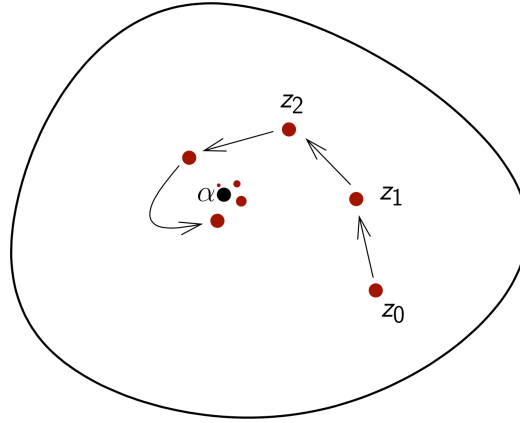


Figure 1: Every root α of f has a neighborhood of initial conditions for which Newton’s method works, i.e., the orbit of z_0 converges to α .

Ernst Schröder in the year 1870 [9, 10] and, independently, Arthur Cayley in the year 1879 [2, 3] were the first to propose a global study of Newton’s method. The questions they posed were of the following kind:

- If f has several roots, which initial conditions have orbits that converge to one or another?
- Do all initial conditions lead to convergent orbits? Do these always converge to some of the roots of f or are there other possible behaviors?

Schröder and Cayley did not get very far in their objectives, but this is how what we know today as complex dynamics was born, which is the local and global study of dynamical systems generated by the iteration of analytical functions in the complex plane.

At this point, it will be useful to define the following concept.

Definition 1.1. Given a root α of the function f , we define its **basin of attraction** as the set of initial conditions of the plane such that their orbit under Newton's method converges to α , i.e.,

$$\mathcal{A}(\alpha) = \{z_0 \in \mathbb{C} \mid N_f^n(z_0) \xrightarrow{n \rightarrow \infty} \alpha\}.$$

Both mathematicians solved the problem for the case in which f is a polynomial of degree 2, proving that each of the two roots of f possesses a half-plane of initial conditions whose orbits converge to the root in question. These two basins of attraction are separated by the perpendicular bisector line between the two roots, formed by initial conditions with orbits that do not converge to either of the two roots, i.e., initial conditions for which the method “fails” (see Figure 2).

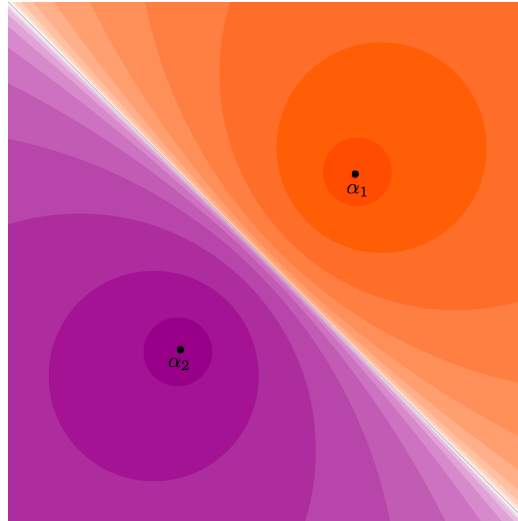


Figure 2: Newton's method for a polynomial P of degree 2 with roots α_1 and α_2 . In orange, the basin of attraction of α_1 is shown, and in purple that of α_2 . The line that separates them is formed by initial conditions for which the method “fails”.

But Cayley gave up regarding degree 3. In his 1880 article [4] he wrote:

... The division [of the plane into different basins] is done without difficulty in the quadratic case; but in the next case, that of the cubic equation, it is not at all obvious what the division is; and the author has not succeeded in finding it ...

The reason Cayley got stuck is that, without knowing it, he was facing a fractal, without any tool to see it! In Figure 3, one can see the three

basins of attraction of Newton's method for a cubic polynomial, each with a different color. This plane of initial conditions is also called the dynamic plane or phase plane.

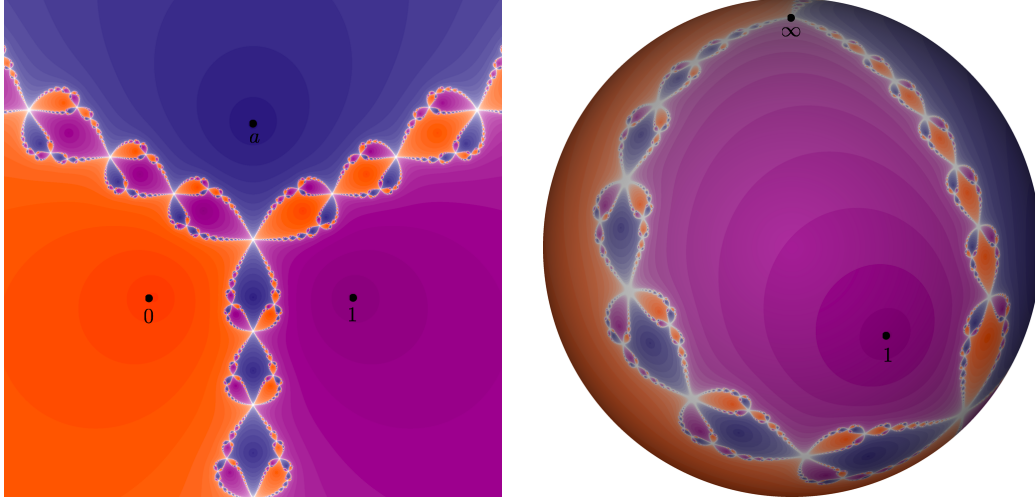


Figure 3: Dynamic space of Newton's method $N_P(z) = z - \frac{P(z)}{P'(z)}$ of the polynomial $P(z) = z(z-1)(z-a)$, with $a = 0.5 + i \sin(\pi/3)$. The initial conditions can be seen in the complex plane and also on the Riemann sphere (the compactification of $\mathbb{C}$ by adding the point at infinity, which corresponds to the north pole). The three basins of attraction of the three roots of P are shown in different colors.

It was not until the year 1918 that Pierre Fatou and Gaston Julia undertook a more systematic study [6, 7, 8] of the iteration of analytical functions, and established the bases of modern theory, of what we know today as complex dynamical systems or complex dynamics.

2 Complex dynamical systems: objectives

We will set aside Newton's method for the moment, and focus on the iteration of more general functions which, for simplicity, we will require to be holomorphic on the Riemann sphere $\hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$, or equivalently, rational functions $F(z) = \frac{P(z)}{Q(z)}$, where P and Q are polynomials without common roots. Given, then, a rational function $F : \hat{\mathbb{C}} \rightarrow \hat{\mathbb{C}}$, our goal is to study the asymptotic behavior of the orbits

$$z_0, z_1 = F(z_0), \dots, z_n = F^n(z_0), \dots$$

as a function of the initial condition z_0 .

Among the simplest orbits we can find, we highlight the following (see Figure 4):

- **Fixed points:** points z_0 such that $F(z_0) = z_0$.
- **Periodic points of period $p > 1$:** points z_0 such that $F^p(z_0) = z_0$ and $F^k(z_0) \neq z_0$ for any $k < p$.
- **Preperiodic points:** points z_0 that are not periodic, but such that $F^k(z_0)$ is periodic for some $k > 0$.



Figure 4: From left to right, a fixed point, a periodic orbit of period $p = 3$, and a preperiodic orbit.

The mentioned orbits are all finite. An infinite orbit can converge, for example, to infinity, or to a fixed point, or it may not converge anywhere. With a slight abuse of language, we will say that the orbit of z converges to a periodic orbit, say $\{z_0, z_1, \dots, z_{p-1}\}$, if $F^{np}(z)$ tends to z_i for some $0 \leq i < p$, as $n \rightarrow \infty$. Note that the exponent is a multiple of p , where p is the length of the periodic orbit.

In studying the phase plane of F , the most basic objective is to classify the initial conditions according to the asymptotic behavior of their orbits. This can be a very simple problem in some cases and, conversely, in others, it can be of very difficult solution or, at present, impossible.

2.1 Invariant sets

One way to simplify the problem we have just posed is to look for what we call invariant sets. The definition of these sets is the following.

Definition 2.1. We say that $A \subset \hat{\mathbb{C}}$ is a set:

- **positively invariant**, if $F(A) \subset A$;
- **negatively invariant**, if $F^{-1}(A) \subset A$, where we understand that

$$F^{-1}(A) = \{z \in \hat{\mathbb{C}} \mid F(z) \in A\};$$

- **totally invariant**, if it is both positively and negatively invariant, or in other words, if $F^{-1}(A) = A = F(A)$.

Invariant sets are interesting in many areas of mathematics but in particular in any dynamical system (real or complex, and of any dimension), since they allow decomposing the system into smaller and perhaps simpler dynamical subsystems to study. In dynamical systems of $\mathbb{R}^n$ we find a multitude of examples such as invariant manifolds, first integrals, etc., which often allow reducing the dimension of the system under study.

The smallest invariant sets are the orbits themselves. Indeed, what we call the positive orbit of z_0 and denote by

$$O^+(z_0) = \{z_0, z_1, \dots, z_n = F^n(z_0), \dots\}$$

is positively invariant. The grand orbit of z_0

$$O(z_0) = \{z \in \hat{\mathbb{C}} \mid F^p(z) = F^q(z_0) \text{ for some } p, q \in \mathbb{N}\}$$

is totally invariant (see Figure 5).

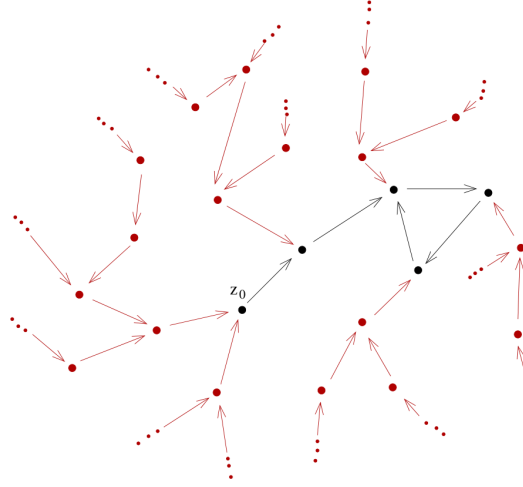


Figure 5: The grand orbit of a point z_0 , $O(z_0)$, consists of the set of points “related” to z_0 through iterates of F . In black we see the positive orbit of z_0 , which in this case is finite. We observe that if z belongs to $O(z_0)$, then $O(z) = O(z_0)$.

Let us return now to examine the dynamical system generated by Newton’s method, and consider what invariant sets we can observe in it.

Recall that if α is a root of a polynomial P , the local study of Newton’s method told us that α is equipped with a neighborhood U of initial conditions whose orbits converge to α . In the previous terminology, this is

saying that U belongs to the basin of attraction of α and, in fact, this basin consists of all preimages (of all orders) of U . That is:

$$\mathcal{A}(\alpha) = \{z_0 \in \mathbb{C} \mid N_p^n(z_0) \xrightarrow{n \rightarrow \infty} \alpha\} = \bigcup_{n \geq 1} N_p^{-n}(U).$$

Observe that U is a positively invariant set, while the entire basin $\mathcal{A}(\alpha)$ is totally invariant. In general, a basin of attraction $\mathcal{A}(\alpha)$ is a set that is:

- open, by continuity;
- totally invariant, by definition, since it is formed by grand orbits, and these are totally invariant;
- possibly formed by infinitely many connected components. It has a main one $\mathcal{A}^*(\alpha)$ which is the one containing α .

In Figure 6, we can see the dynamic plane of Newton's method for another cubic polynomial. We observe three basins of attraction in three different colors. Apparently (and in fact, it is so) one of these is formed by a single connected component, while the others have infinitely many.

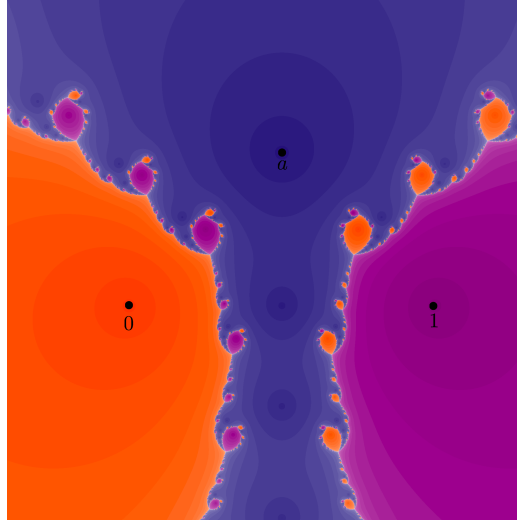


Figure 6: Phase plane of Newton's method N_p for $P(z) = z(z-1)(z-a)$ with $a = \frac{1}{2}(1+i)$. We see three totally invariant basins of attraction. One of these, that of a , is formed by a single connected component, while the other two have infinitely many.

Observe at this point that, although we defined the concept of basin of attraction for Newton's method, this is easily generalizable to other functions. Given a rational function F with a fixed (or periodic) point that attracts the orbits of a neighborhood (points with this property are called

attractors), its basin of attraction is, analogously, the set of orbits that converge to it. Thus defined, any basin of attraction has all the mentioned properties: it is open, totally invariant, and possibly formed by infinitely many connected components.

It is not difficult to check that if a set A is totally invariant, its complement and its boundary must also be totally invariant. Therefore, we obtain the second example of invariant sets: if $\mathcal{A}(\alpha)$ is a basin of attraction, its boundary is also totally invariant. By definition, this set is closed and cannot have an interior.

3 The basic partition: the Julia and Fatou sets

There is a fundamental difference between the two examples of totally invariant sets we have presented: the basins of attraction belong to what we call the Fatou set or stable set, while their boundaries belong to the Julia set or chaotic set. Let us see now how these two sets are defined, which are in more than one aspect complementary to each other. We previously need the concept of normality of the family of iterates in a certain open set of the plane.

Definition 3.1. Let $F : \hat{\mathbb{C}} \rightarrow \hat{\mathbb{C}}$. We say that the family of iterates $\{F^n\}_{n \in \mathbb{N}}$ is **normal** in an open set $U \subset \hat{\mathbb{C}}$ if every subsequence of $\{F^n\}_{n \in \mathbb{N}}$ has a sub-subsequence that converges uniformly on compact subsets of U (possibly to ∞).

Informally, the family of iterates is normal in an open set when all points of the open set behave similarly under the iteration of F , which is why we relate this concept to stability.

Definition 3.2. The **Fatou set**, $\mathcal{F}(F)$, is defined as the set of points $z \in \hat{\mathbb{C}}$ for which the family of iterates $\{F^n\}_{n \in \mathbb{N}}$ is normal in some neighborhood U of z . Its complement is called the **Julia set**. That is, $\mathcal{J}(F) = \hat{\mathbb{C}} \setminus \mathcal{F}(F)$ is the set of points for which the family of iterates is not normal in any of its neighborhoods.

From the definition itself, it is not difficult to deduce that any basin of attraction of a fixed point α belongs to the Fatou set, since the sequence of iterates of F itself converges uniformly on compact sets to the constant function $G(z) \equiv \alpha$. The boundary of $\mathcal{A} = \mathcal{A}(\alpha)$, on the other hand, belongs to the Julia set. Let us give a short proof of this.

Proof. Indeed, if z_0 belongs to $\partial\mathcal{A}$, any neighborhood U of z_0 contains points of $\mathcal{A}$ and of its complement. Any limit function G of some subsequence of iterates would have to be equal to α for all $z \in U \cap \mathcal{A}$, which is a non-empty open set. Since G is analytic (by Weierstrass's theorem), the identity theorem (or analytic continuation principle) would tell us that G must be identically equal to α on the whole of U , which contradicts the fact that no subsequence of iterates of z_0 can converge to α , because z_0 does not belong to the basin of attraction. $\square$

From the definition itself, we can see that the Julia and Fatou sets are totally invariant, since the concept of normality is an asymptotic concept associated with grand orbits. We will see next that these two sets are completely opposite, both topologically and dynamically. For this reason, we say that they form the basic partition of the phase space. Figure 7 shows phase planes of different functions where we can see examples of Fatou and Julia sets.

3.1 Topological properties of the Julia set

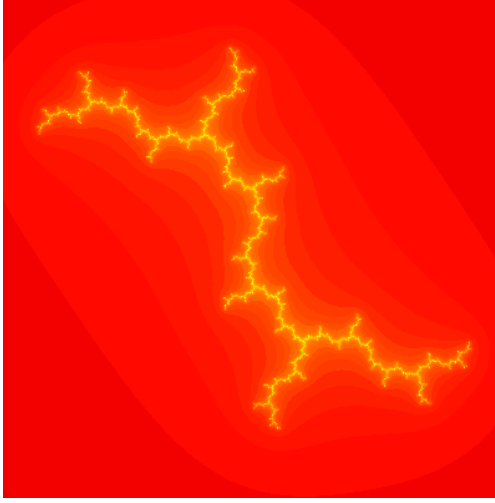
Next, we will state several topological properties of the Julia set of a rational function. Some are simple to prove, while others require proofs beyond the scope of this text.

Proposition 3.1. *Let $F : \hat{\mathbb{C}} \rightarrow \hat{\mathbb{C}}$ be a holomorphic function and let $\mathcal{J}$ be its Julia set. Then:*

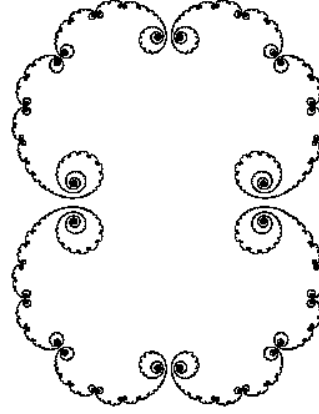
- $\mathcal{J}$ is closed and non-empty.
- $\text{int}(\mathcal{J}) = \emptyset$ or else $\mathcal{J} = \hat{\mathbb{C}}$.
- $\mathcal{J}$ is “self-similar”, that is, in all its parts we find scaled copies of the original set.

There are examples of topologically very different Julia sets, as we have begun to see in Figure 7. As a small sample, we mention that there exist examples of different Julia sets that are:

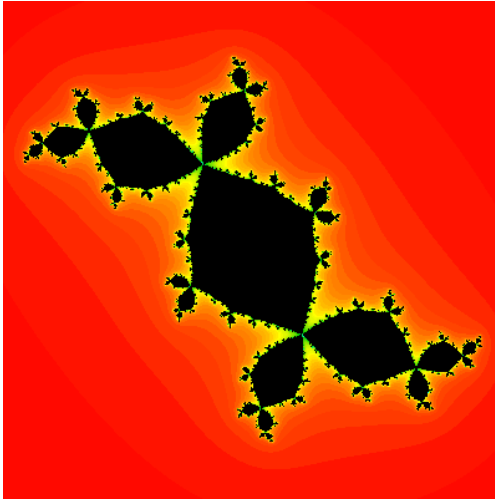
- connected, disconnected, or Cantor sets;
- locally connected and non-locally connected;
- with Lebesgue measure zero and with positive measure (but without interior!);



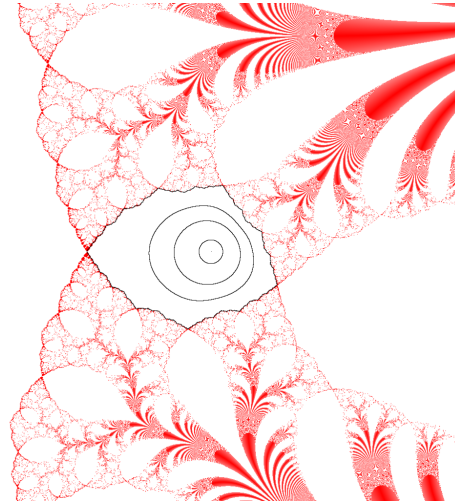
(a) $F(z) = z^2 + i$.



(b) $F(z) = z^2 + 0.25$.



(c) $F(z) = z^2 - 0.122\dots + 0.744\dots i$.



(d) $F(z) = e^{2\pi i \theta} z e^z$, $\theta = \frac{1-\sqrt{5}}{2}$.

Figure 7: a) In red we see the basin of attraction of infinity which coincides with $\mathcal{F}(F)$. The Julia set (yellow) is a dendrite (closed, connected, and with no interior). b) In white we see $\mathcal{F}(F) = \mathcal{A}(\infty)$. The Julia set (black) is a Cantor set. c) In red we see $\mathcal{A}(\infty)$ and in black its complement K , known as “Douady’s rabbit”. The common boundary to these two sets is $\mathcal{J}(F)$. The points in the interior of K form part of the basin of attraction of a periodic orbit of period 3. d) In red we see $\mathcal{J}(F)$, which consists of a set of “filaments”. In white we see $\mathcal{F}(F)$ and in black some of its orbits, all accumulating on invariant curves, where F is basically a rotation of irrational angle.

- with Hausdorff dimension $1 + t$ for all $t \in (0, 1]$.

It is worth noting that the existence of rational functions with Julia sets of positive measure, but without interior, was a highly disputed conjecture for more than twenty years, which was resolved positively very recently by X. Buff and A. Chéritat (2005). These authors showed that even within the simplest families (quadratic polynomials) one can find Julia sets with this property.

3.2 Dynamical properties of the Julia sets

We could almost say that, in a natural way, the dynamics on the Julia set is unstable, since it is the boundary between different stable behaviors and, therefore, must reflect the tension between them. In fact, the orbits of the Julia set exhibit all the properties usually associated with the concept of chaos. Without defining what it means for a function to be chaotic (which is actually just a matter of convention), we will list some of these characteristics, without proof. The dynamics of the orbits of the Julia set thus exhibit the following properties:

1. $F|_{\mathcal{J}}$ is sensitive with respect to initial conditions. Intuitively, this means that orbits that start very close to each other eventually diverge. Formalizing, there exists $k > 0$ such that for every $z_0 \in \mathcal{J}$ and for every $\epsilon > 0$, there exists $z_1 \in \mathcal{J}$ with $|z_0 - z_1| < \epsilon$ and there exists $n > 0$ such that $|F^n(z_0) - F^n(z_1)| \geq k$.
2. The (grand) orbits are dense in the whole of $\mathcal{J}$. For any point $z \in \mathcal{J}$, the grand orbit of z is dense in $\mathcal{J}$. In fact, the backward orbit $O^-(z)$ (i.e., the set of preimages of z) is also dense in $\mathcal{J}$.
3. F is transitive on $\mathcal{J}$. Informally, this means that F “mixes” all orbits. Formally, for any neighborhoods U, V of points in $\mathcal{J}$, we have $F^n(U) \cap V \neq \emptyset$ for some n .
4. The (non-attracting) periodic points of F are dense in $\mathcal{J}$. In other words, we find periodic points as close as we want to any point of $\mathcal{J}$. Note that they cannot be attracting, since these form part of the stable set.

Note that from the previous properties, for example from the second or the third, the following fact is deduced.

Proposition 3.2. *$\mathcal{J}$ cannot have any proper totally invariant subset. That is, there does not exist any non-empty, totally invariant subset $A \subset \mathcal{J}$ such that $\overline{A} \neq \mathcal{J}$.*

In other words, we can say that the Julia set can no longer be decomposed further in the dynamical sense, that is, into significant totally invariant parts. This is not the case for its complement, as we will see next.

4 The Fatou or stable set

Recall that the Fatou set $\mathcal{F}$, the complement of $\mathcal{J}$, is open and contains all initial conditions with orbits showing stable asymptotic behavior. In fact, we have already seen an example: the basin of attraction of an attracting fixed point is always part of $\mathcal{F}$. The condition for a fixed point to be attracting is that $|F'(z_0)| < 1$, which can be easily checked by looking at the first terms of the Taylor expansion of F near z_0 .

It is natural to ask, however, what other behaviors are possible for the orbits in $\mathcal{F}$. Below we see some examples.

Example 1. Periodic basin of attraction. The orbits “converge” to an attracting periodic orbit, in the sense defined above. Generalizing the case of a fixed point, given a periodic orbit $\{z_0, z_1, \dots, z_{p-1}\}$ of period p , the necessary and sufficient condition for it to be attracting is that its multiplier has modulus less than 1, i.e.:

$$|\lambda| = |(F^p)'(z_i)| = |F'(z_0)F'(z_1) \cdots F'(z_{p-1})| < 1.$$

In Figure 8 we see the phase plane of the function $F(z) = z^2 - 1$.

The orbit $\{0, -1\}$ is an attracting period-2 orbit. This is surrounded by an immediate basin of attraction that corresponds to the two yellow (open) components containing it. The union of all preimages of these forms the totality of the basin drawn in yellow. Its complement, drawn in shades of red, is another basin of attraction, in this case of the point at infinity. The common boundary between the two basins, both of the stable set, is precisely the Julia set (in black).

Example 2. Parabolic basin. Another example of stable behavior is found by considering what we call parabolic (attraction) basins. They are very similar to the previous ones, but with the difference that the “limit” points (fixed or periodic) are on the boundary of the basin, and not in its interior.

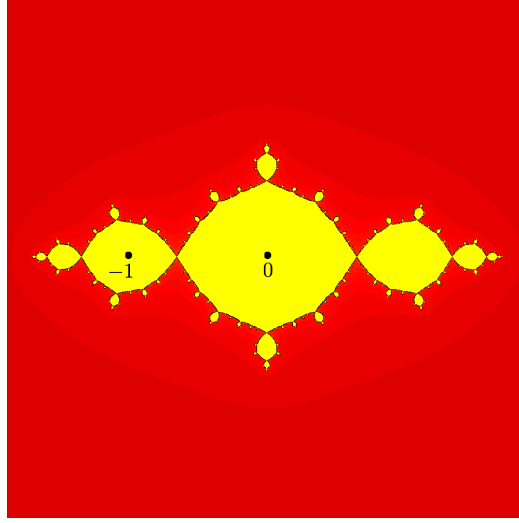


Figure 8: The phase plane of $F(z) = z^2 - 1$. The orbit $\{0, -1\}$ is 2-periodic and attracting. In yellow we see its basin of attraction, part of the Fatou set. In red we see the basin of attraction of the point at infinity. The common boundary of both basins is the Julia set.

As before, we also have a necessary and sufficient condition for a periodic orbit $\{z_0, z_1, \dots, z_{p-1}\}$ of period p to be parabolic, and this is that its multiplier has modulus exactly 1, with the property

$$|\lambda| = |(F^p)'(z_0)| = e^{2\pi i p/q}, \quad p, q \in \mathbb{Z}, \quad \gcd(p, q) = 1.$$

In Figure 9 we can see the phase plane of the polynomial $F(z) = z^2 + 0.25$, which has a fixed point at $z_0 = 1/2$, with multiplier $\lambda = 1$.

In the figure, we see in yellow its parabolic basin of attraction, in this case formed by a single connected component. All points of this color have orbits that converge to z_0 , located on the boundary. As before, in red we see the basin of attraction of infinity and, in black, the common boundary that corresponds to the Julia set (in this case a Jordan curve, not differentiable at any of its points).

Example 3. Rotation disk. A very different example of a stable component is what is called a rotation disk, also known as a Siegel disk in honor of Carl Siegel (1896–1981), who gave sufficient conditions for its existence.

What is known as a fixed rotation disk for F is an open set U conformally equivalent to the unit disk, which contains a fixed point z_0 in its interior, and such that $F|_U$ is conformally conjugate to a rigid rotation

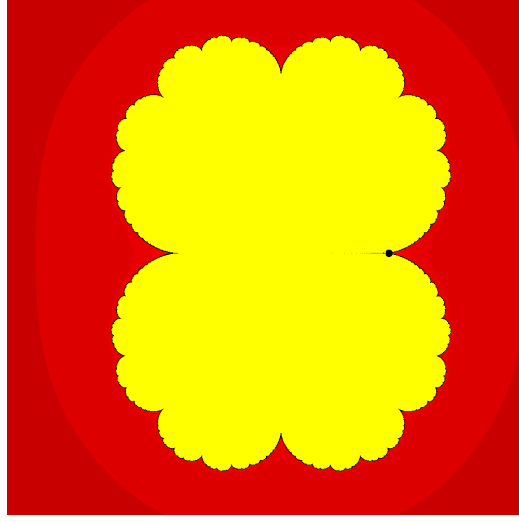


Figure 9: Phase plane of $F(z) = z^2 + 0.25$, for which $z_0 = 1/2$ is a parabolic fixed point. In yellow, its parabolic basin is shown, which has z_0 on its boundary (right). In red, the basin of attraction of infinity.

$\mathcal{R}_\theta(z) = e^{2\pi i\theta}z$ on the unit disk, where $\theta \in \mathbb{R} \setminus \mathbb{Q}$. This means that there exists a conformal homeomorphism $h : \mathcal{U} \rightarrow \mathbb{D}$ such that $h \circ F = \mathcal{R}_\theta \circ h$.

This means that the orbits inside $\mathcal{U}$ lie on invariant curves and, on these, they behave like an irrational rotation, i.e., they accumulate densely over the entire curve.

In Figure 10 we show the phase plane of the function $F(z) = \lambda z(1 - z)$ where $\lambda = e^{2\pi i \frac{1-\sqrt{5}}{2}}$.

The black point corresponds to a fixed point and the curves around it are some of the invariant curves. The orbit of any of its points is dense on the curve. The yellow component containing these elements is the fixed Siegel disk $\mathcal{U}$. All other components of the same color are preimages of $\mathcal{U}$ and together they form the grand orbit of $\mathcal{U}$, which is totally invariant. In red we see again the basin of attraction of infinity and in black the common boundary which is the Julia set. In this case, we also have sufficient conditions that ensure the existence of this type of stable components. Siegel already proved that if z_0 is a fixed point with multiplier $\lambda = F'(z_0) = e^{2\pi i\theta}$, with θ a Diophantine irrational number, then there is always a Siegel disk around z_0 . More recently (1971), this condition was improved by Brjuno [1], by proving the same fact for all numbers θ in the Brjuno set, a set of irrationals that includes Diophantine numbers. Finding necessary conditions for the existence of rotation disks is a more difficult problem that depends on the specific family being studied. In fact, in most cases, this is

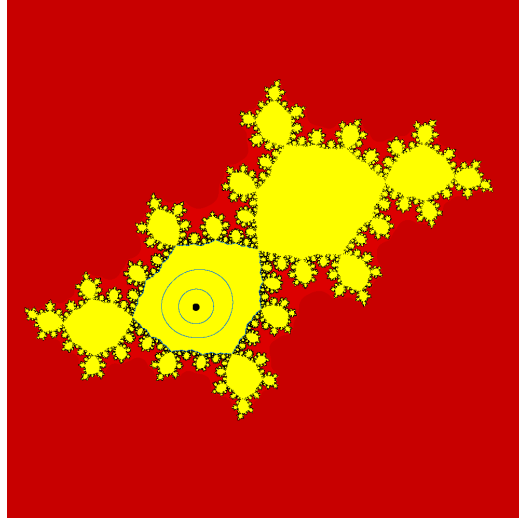


Figure 10: Phase plane of $F(z) = \lambda z(1 - z)$ where $\lambda = e^{2\pi i \frac{1-\sqrt{5}}{2}}$. In yellow a fixed Siegel disk around $z = 0$ and all its preimages are shown. In red, the basin of attraction of infinity.

still an open problem.

Example 4. Rotation ring. This is an example very similar to the previous one, with the only difference that the domain is conformally equivalent to an annulus, instead of a disk. In this annulus, we have the same phenomenon as before, i.e., the function is conjugate to an irrational rotation and, therefore, all orbits are dense on invariant curves. These rings are also known as Herman rings in honor of Michel Herman (1942–2000), who found the first examples.

Rotation rings differ from the previous examples in that they are not associated with any periodic orbit. Because of this, we do not have any simple criterion from which their existence can be deduced for a specific function.

As a consequence of the maximum modulus principle, it is established that any rotation ring requires the presence of a pole of the function. For this reason, we can never find rotation rings in the phase planes of polynomials or entire functions in general.

In Figure 11 we can see the phase plane of the rational function $F(z) = z^2 \frac{z-a}{1-\bar{a}z}$, for a certain conveniently chosen value of a . In gray, a Herman ring is shown, inside of which one of its invariant curves is drawn. All white ring-shaped components are preimages (of different orders) of the

main ring. The disk-like white components form part of the basin of infinity and the basin of zero, which in this case are both, zero and infinity, attracting fixed points.

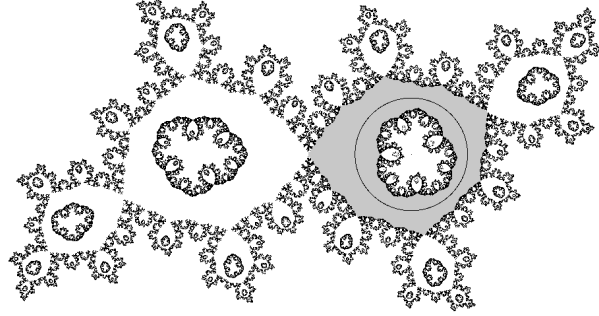


Figure 11: Phase plane of $F(z) = z^2 \frac{z-a}{1-az}$ for a certain conveniently chosen value of a . In gray, a Herman ring is shown, inside of which one of its invariant curves is drawn.

Example 5. Parabolic domain at infinity. A fixed parabolic domain at infinity is an open set U of the plane such that all its orbits converge toward the point at infinity, which must be an essential singularity. These components are also known by the name of Baker domains in honor of Noel Baker (1932–2001), who made the first exhaustive studies of them.

By definition, then, these stable components only exist in transcendental functions (i.e., with essential singularities). We find ourselves again with a type of components that is not associated with any periodic orbit either.

In Figure 12 we see the phase plane of $F(z) = z + a + b \sin(z)$ with a and b chosen appropriately so that the Fatou set has a Baker domain, shown in red. The orbits inside this set escape to the right, tending toward infinity, with an essentially linear velocity. In blue, the preimages of the Baker domain can be seen.

Example 6. Wandering domains. A wandering domain is an open set U such that $F^n(U) \cap F^m(U) = \emptyset$ for all $n, m \in \mathbb{N}$ with $n \neq m$. That is, wandering domains are neither periodic nor preperiodic components.

In Figure 13 we see the dynamic plane of the transcendental function $F(z) = z + 2\pi + \sin(z)$. The central components colored gray, centered at the points $z_k = (2k + 1)\pi$, correspond to an orbit of wandering domains. The smaller ones are preimages of these.

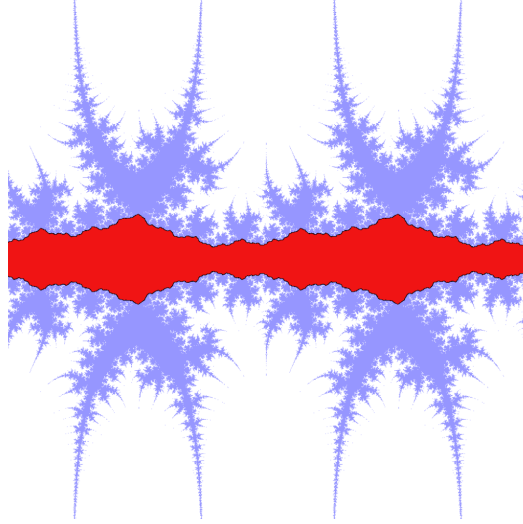


Figure 12: Phase plane of $F(z) = z + a + b \sin(z)$ with a and b chosen appropriately so that the Fatou set has a Baker domain, shown in red. Orbits inside this domain tend toward infinity with linear velocity. In blue we see all preimages of the Baker domain.

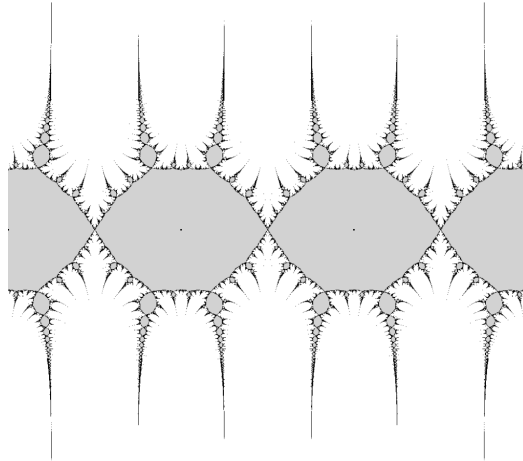


Figure 13: Phase plane of the transcendental function $F(z) = z + 2\pi + \sin(z)$. The central components colored gray correspond to an orbit of wandering domains.

Fatou and Julia proved that, in theory, these domains were possible. It was not until much later, in 1985, that Dennis Sullivan proved the celebrated theorem establishing that rational functions cannot have wandering domains. This result opened a new era of complex dynamics, not because of the statement itself, but because of the techniques used in its proof, which were highly innovative at that time.

Theorem 4.1 (Sullivan, [11]). *Rational functions $(F(z) = \frac{Q_1(z)}{Q_2(z)})$, where Q_1, Q_2 are polynomials) do not have wandering domains.*

4.1 Classification of stable components

The previous examples have shown us a variety of components of the Fatou set, where each exhibits a different behavior of the iterates of the function. Fatou, in 1918, did not have the possibility to “see” these examples, but nevertheless, he was able to prove the following classification theorem.

Theorem 4.2 (Fatou, 1918). *Let U be a periodic connected component of $\mathcal{F}(F)$ where F is a rational function on $\hat{\mathbb{C}}$. Then U is part of a basin of attraction, of a parabolic basin, of a rotation disk, or of a rotation ring.*

If F is transcendental, U can also be a parabolic domain at infinity. If U is not periodic, then U is a preimage of some of the previous ones or it can be a wandering domain.

Fatou found examples of most of these behaviors, and conjectured that probably rotation domains did not exist. It was not until many years later that their existence could be proven, as we have mentioned before, as well as the non-existence of wandering domains for rational functions.

4.2 Newton’s method revisited

We had seen that, given a polynomial $P(z)$ with roots $\alpha_1, \dots, \alpha_k$, each is equipped with its basin of attraction under Newton’s method $F(z) = N_P(z)$, consisting of all initial conditions with orbits that converge toward the root in question. In fact, it is simple to check that $N_P(\alpha_i) = \alpha_i$ and that $N'_P(\alpha_i) = 0$, if α_i is a simple root, but in any case, $|N'_P(\alpha_i)| < 1$. This tells us that the roots of P are attracting fixed points of N_P and that, therefore, they have a basin of attraction.

$$\mathcal{A} = \mathcal{A}(\alpha_1) \cup \dots \cup \mathcal{A}(\alpha_k).$$

But in the dynamic plane of N_P (the space of initial conditions), what else is there besides the basins of attraction? Which initial conditions have orbits that do NOT converge to any of the roots of P ? We will call these initial conditions **bad**, in the sense that they do not work for the purpose of Newton's method.

clearly, all initial conditions in the Julia set are bad. But in most cases,¹ the measure of the Julia set is zero and, therefore, the probability of starting randomly with an initial condition in $\mathcal{J}$ is nil. But, in addition, all initial conditions in

$$\mathcal{F}(N_P) \setminus \mathcal{A},$$

which is open and not necessarily empty, are also bad. How can we know if it contains points? For a fixed polynomial it is simple to perform a numerical experiment that indicates whether this set is non-empty. As we have done in previous images, in Figure 14 we have colored points of a grid with different colors depending on whether their orbits tend to one root or another (and with more or less intensity depending on the speed with which they do so), or in black if they do not converge to any of them.

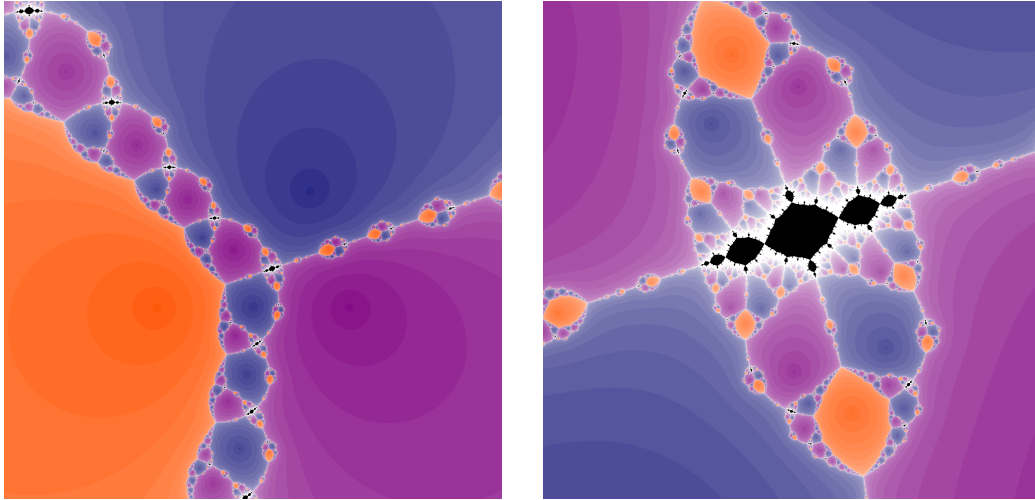


Figure 14: **Left:** the phase plane of Newton's method of the polynomial $P(z) = z(z - 1)(z - \alpha)$ with α chosen appropriately. **Right:** a zoom in which we see one of the sets of bad initial conditions. In this concrete case, it consists of the basin of attraction of a periodic orbit of period 2.

We can expect not to see the initial conditions of the Julia set in black,

¹It could be quantified by saying that the Julia set of Newton's method has measure zero for almost every fixed polynomial of degree $k > 0$, considering the space of functions and the appropriate measure.

since having measure zero, they will certainly not be on the computation grid. Even so, we will see them almost white thanks to the gradation of colors, since the closer an initial condition is to $\mathcal{J}$, the longer its orbit will take to approach the corresponding root. In Figures 3, 6, and 14, we can see that no black set is observed, which indicates to us that probably the two polynomials chosen in those cases do not have bad initial conditions for Newton's method, apart from those of the Julia set. But a different case is seen with the polynomial of Figure 14, where the existence of sets with positive measure that do not belong to any basin of attraction is obvious. With any enlargement, these sets of bad initial conditions become perfectly visible.

Once their existence is established, it is natural that we ask ourselves how many of these sets we can find in a dynamic plane. The probable answer is that there are infinitely many of them, since normally stable components have infinitely many preimages. But it would be interesting to know how many "different" ones we can find. In the example of Figure 14 we see an infinity of black "spots". Taking some of them at random and with a few calculations we could check that they all correspond to a basin of attraction of a periodic orbit of period 2. In other words, it seems that all bad initial conditions, apart from those in $\mathcal{J}$, have orbits that converge to the same attracting periodic orbit. That is, it seems that they all belong to the same basin of attraction. But how do we know this for sure?

4.3 Entire Fatou components and singularities of the inverse

We will say that $\mathcal{U}$ is a **grand Fatou component** if it is the grand orbit of a connected component of the Fatou set. When we consider a grand orbit, we have an open, totally invariant set, typically formed by infinitely many connected components.

The problem of the previous section can be reformulated in general and in terms of grand Fatou components. Given a specific function F , how do we know which grand Fatou components are in its dynamic plane? How many different ones can there be? To answer this question, we must talk about the singularities of F^{-1} . For simplicity, we will assume, as we have done so far, that F is a rational function, as would be the case for Newton's method of a polynomial.

Any rational function $F(z) = \frac{Q_1(z)}{Q_2(z)}$ is a continuous and analytic function at all points when we consider it as a function on the Riemann sphere, i.e., on $\hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$. In addition, F is a local homeomorphism in a neighbor-

hood of all its points, except in neighborhoods of the zeros of the derivative F' . This motivates the following definition.

Definition 4.1. We say that c is a **critical point** of F if $F'(c) = 0$. Then, $v = F(c)$ is called a **critical value** and is a singularity of F^{-1} , in the sense that the branch of F^{-1} that maps v to c is not well-defined in any neighborhood of v (inverse function theorem).

Since F is a quotient of polynomials, it only has a finite number of critical points, $2d - 2$ to be precise, where d is the maximum degree between Q_1 and Q_2 . These points play a crucial role in the dynamics of F , as shown by the following theorem.

Theorem 4.3 ([7, 8]). *If $\mathcal{U} \subset \mathcal{F}(F)$ is a grand Fatou component, then $\mathcal{U}$ has at least one associated critical point. More specifically:*

- *Basins of attraction and parabolic basins contain at least one critical point.*
- *Siegel disks and Herman rings have the orbit of some critical point that accumulates on their boundary (in fact, two for the ring!).*

We want to emphasize that this fact is very particular to analytic functions of one variable. Nothing similar is true for differentiable dynamical systems, nor for analytic ones looked at only on the real line, except in very special cases.

Observe that this provides us with great control over the stable sets. In some way, we only need to iterate the critical points to locate all possible stable components.

5 Newton's method and its parameter plane

Finally, we would like to apply again the general concepts we have seen to Newton's method. For the previous theorem, it is important to see first what the critical points of N_P are and what their orbits are like.

If P is a polynomial of degree $k > 2$, the critical points of Newton's method $N_P(z) = z - \frac{P(z)}{P'(z)}$ are the solutions of

$$N'_P(z) = \frac{P(z)P''(z)}{(P'(z))^2} = 0.$$

Therefore (assuming simple roots of P):

$$\{\text{Critical points of } N_P\} = \{\text{Zeros of } P\} \cup \{\text{Zeros of } P''\}.$$

The former are the roots of P that we have called $\alpha_1, \dots, \alpha_k$ and which are already the critical points associated with their respective basins of attraction.

But the latter are the inflection points of P (there are $k - 2$ of them) and these remain free to generate, or not, other Fatou components. In other words, Newton's method N_P can have at most $k - 2$ grand Fatou components different from the basins of attraction of the roots of P , one for each free critical point.

Let us study in more detail the special case of polynomials of degree 3. For a cubic polynomial, N_P has four critical points: the three roots of P , $\alpha_1, \alpha_2, \alpha_3$, and a free critical point, let us call it c . As we have seen, the behavior of the orbit of the critical point, $O^+(c)$, gives us information about the possible grand Fatou components that we can find in the dynamic plane, different from the basins of attraction $\mathcal{A}(\alpha_1), \mathcal{A}(\alpha_2), \mathcal{A}(\alpha_3)$, the union of which we have called $\mathcal{A}$. For example, in the case where the critical point c is attracted by one of the roots, it can no longer be generating any other stable component. This gives us a sufficient condition, though not necessary, for a specific Newton's method to work at almost every point of the plane (assuming that the Julia set has measure zero):

$$O^+(c) \rightarrow \alpha_i \text{ for some } i \in \{1, 2, 3\} \Rightarrow \mathcal{F}(N_P) = \mathcal{A}$$

and, therefore, the only "bad" initial conditions are found in $\mathcal{J}$. When this is not the case, there is the possibility (not the obligation) that other grand Fatou components exist.

This condition can be checked by Newton's method for each polynomial in particular. It is a fact, not difficult to verify, that every cubic polynomial is conjugate by an affine map to another of the form

$$P_a(z) = z(z - 1)(z - a).$$

It is, therefore, sufficient to study the family P_a to know how the Newton's methods of all polynomials of degree 3 behave. For this family, once the value of a is fixed, we would have to iterate the free critical point $c_a = \frac{a+1}{3}$ (the zero of P''). We can show the result in the complex plane, where each point of the plane represents a value of the parameter a , and where we paint the point with different colors depending on whether the orbit of c_a is attracted by some of the roots (colors blue, purple, or orange) or is not attracted by any (black).

In this figure, we can then observe a multitude of black spots, apparently open sets of parameter values, for which Newton's methods of the corresponding polynomials possibly have sets of positive measure of bad

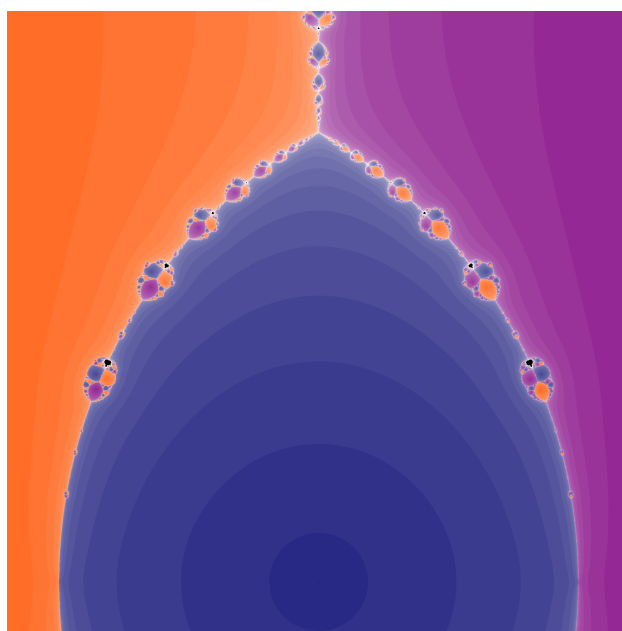


Figure 15: Parameter plane of Newton's method for the family of cubic polynomials $P_a(z) = z(z-1)(z-a)$. Each point of the plane represents a value of a and is colored if the orbit of the free critical point is attracted by some of the roots 0 , 1 , or a , and is black if it has any other behavior.

initial conditions. In Figure 16 (left), a zoom is shown around one of these black spots, where we see that the spot in question does not have just any shape. Quite the contrary, the set we see is a (quasiconformal) copy of the celebrated Mandelbrot set M , the bifurcation set of the quadratic family $Q_c(z) = z^2 + c$, as shown in Figure 16 (right). This is the parameter plane of the polynomials Q_c (not of their Newton's methods!), and what we see is the result of iterating their critical point $z = 0$ for the different polynomials Q_c , coloring the values of c according to the behavior of this orbit.

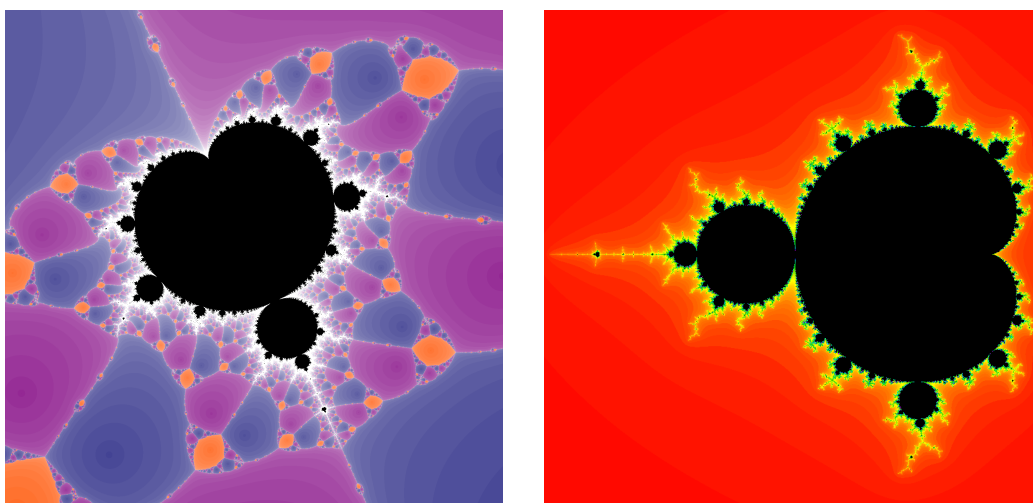


Figure 16: **Left:** zoom around one of the small black spots of Figure 15. **Right:** the Mandelbrot set, in the parameter space of quadratic polynomials $Q_c(z) = z^2 + c$ (not of their Newton's methods!). In other words, the result of iterating the critical point $z = 0$ for each of the polynomials Q_c .

The “unexpected” appearance of the Mandelbrot set in a parameter space foreign to the family Q_c is neither accidental nor strange. We know it is not accidental, as it was explained in detail in 1985 by means of the theory of polynomial-like mappings by Douady and Hubbard [5]. This theory can be summarized by saying that functions of all kinds, under certain conditions, behave locally like polynomials. This is why we see polynomial Julia sets in their dynamic planes (see Figure 14 and compare it with Figure 8), and Mandelbrot sets (or other analogue ones of higher degree) in their parameter spaces. From this relationship, much information is deduced about the dynamics for the parameters that live inside these copies of M . For example, we will highlight one fact: for all parameters in the interior of a copy of M , the corresponding dynamic planes have an attracting periodic orbit and, therefore, a basin of attraction, or else the

Julia sets have positive measure. We deduce, therefore, that we have open sets of parameters (all the interiors of the copies of M) for which Newton's methods fail on sets of positive measure.

We will end this lecture by remarking that finding copies of the Mandelbrot set in various parameter spaces is not at all strange, since the conditions required for this to happen are relatively common and, indeed, occur in a multitude of families of all kinds; not only rational like those that have occupied us, but also transcendental. Because of this fact, we say that the Mandelbrot set is a universal set. And this is also why the Mandelbrot set has been and continues to be an object of great interest to a multitude of mathematicians, especially also because it is an inexhaustible source of open problems, many of which have resisted being solved to this very day.

References

- [1] BRJUNO, A. D. "Analytic form of differential equations." *Trans. Mosc. Math. Soc.*, 25 (1971).
- [2] CAYLEY, A. "Applications of the Newton-Fourier Method to an imaginary root of an equation." *Quat. J. of Pure and App. Math.*, 16 (1879), 179–185.
- [3] CAYLEY, A. "The Newton-Fourier Method imaginary problem." *Am. J. of Math.*, 2 (1879), 97.
- [4] CAYLEY, A. "On the Newton-Fourier imaginary problem." *Proc. Camb. Phil. Soc.*, 3 (1880), 231–232.
- [5] DOUADY, A.; HUBBARD, J. "On the dynamics of polynomial-like mappings." *Ann. Scient. Ec. Norm. Sup.*, 18 (1985), 287–343.
- [6] FATOU, P. "Sur les équations fonctionnelles." *Bull. Soc. mat. Fr.*, 47 (1919), 161–271; 48 (1920), 33–94, 208–314.
- [7] FATOU, P. "Sur les substitutions rationnelles." *Comp. Rend. heb. S. Acad. Sci.*, 164 (1917), 806–808; 165 (1917), 992–995.
- [8] JULIA, G. "Mémoire sur l'iteration des fonctions rationnelles." *J. Math. Pures Appl.*, 8 (1918), 47–245.
- [9] SCHRÖDER, E. "Ueber unendlich viele Algorithmen zur Auflösung der Gleichungen." *Mathematische Annalen*, 2 (1870), 317–365.

- [10] SCHRÖDER, E. "Ueber iterirte Functionen." *Mathematische Annalen*, 3 (1871), 296–322.
- [11] SULLIVAN, D. "Quasiconformal homeomorphisms and dynamics." *Annals of Math.*, 122 (1985), 401–418.